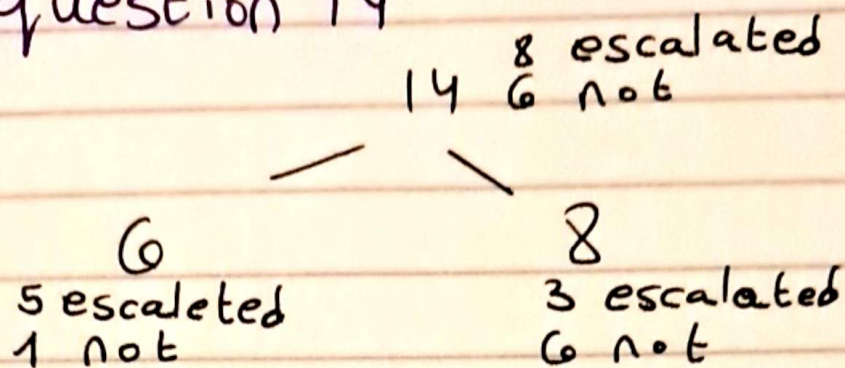


question 14



a) $Gini(Parent) = 1 - \left[\left(\frac{6}{14} \right)^2 + \left(\frac{8}{14} \right)^2 \right] = 1 - 0.51 = 0.49$

$Gini(P) = 0.49$

b)

$Gini(Left) = 1 - \left[\left(\frac{5}{6} \right)^2 + \left(\frac{1}{6} \right)^2 \right] = 1 - 0.722 = 0.277$

$Gini(L) = 0.277$

$Gini(Right) = 1 - \left[\left(\frac{3}{8} \right)^2 + \left(\frac{5}{8} \right)^2 \right] = 1 - 0.53 = 0.46$

$Gini(R) = 0.46$

c)

$G_{split} = \frac{6}{14} \cdot 0.277 + \frac{8}{14} \cdot 0.46 = 0.38$

$G(split) = 0.38$

$Gain = 0.49 - 0.38 = 0.11$

$Gain = 0.11$

d

$$0.08 < 0.11$$

בשאלה הייתה ה Gain קטן
למר. כל של Gain גדול יותר ננו
העצבים טוב יותר כיוון של Gain
מרב כמה חוסר טובה העצבים לחסון
לכן השאלה היא מה למר היא
השאלה ששאלה בשאלה.

"does the ticket contain....."

question 15

(a) $\bar{y} = \frac{5 + 7 + 15 + 17 + 19 + 21}{6} = 14$

y	5	7	15	17	19	21	sum
y - $\bar{y}$	-9	-7	1	3	5	7	0
(y - $\bar{y}$) ²	81	49	1	9	25	49	214

$$SSE = 81 + 49 + 1 + 9 + 25 + 49 = 214$$

$$MSE = \frac{1}{6} \cdot 214 = 35.66$$

$$MSE = 35.66$$

$$\text{issues} \leq 3$$

$$\textcircled{6} \quad \begin{array}{cc} 6 & - \\ 1, 2 & 3, 4, 5, 6 \end{array}$$

Yes: $\bar{y} = \frac{5+7}{2} = 6$

y	S	7	Sum
y - $\bar{y}$	-1	1	0
(y - $\bar{y}$) ²	1	1	2

$SSE(\text{yes}) = 2$

$MSE = \frac{2}{2} = 1$

No: $\bar{y} = \frac{15+17+19+21}{4} = 18$

y	15	17	19	21	Sum
y - $\bar{y}$	-3	-1	1	3	0
(y - $\bar{y}$) ²	9	1	1	9	20

$SSE = 20$

$MSE = \frac{1}{4} \cdot 20 \cdot S$

$MSE_{\text{split}} = \frac{2}{6} \cdot 1 + \frac{4}{6} \cdot S = 3.66$

$\text{Gain} = 35.66 - 3.66 \cdot 4$

© Premium ?

no. $\begin{matrix} & 6 \\ 1, 3 & & 2, 4, 5, 6 \end{matrix}$

No. $\bar{y} = \frac{5 \cdot 15}{2} = 10$

y	S	15	Sum
$y - \bar{y}$	-5	5	0
$(y - \bar{y})^2$	25	25	50

$SSE = 50$

$MSE = \frac{1}{2} \cdot 50 = 25$

Yes: $\bar{y} = \frac{7 + 17 + 19 + 21}{4} = 16 = \bar{y}$

y	7	17	19	21	Sum
$y - \bar{y}$	-9	1	3	5	0
$(y - \bar{y})^2$	81	1	9	25	116

$SSE = 116$

$MSE = \frac{1}{4} \cdot 116 = 29$

$MSE_{split} = \frac{2}{6} \cdot 25 + \frac{4}{6} \cdot 29 = 27.66$

$Gain = 35.66 - 27.66 \cdot 8$

(d) Split B at Sale. 46.7
 A se Gain = 4 1N185 8 1 se Gain 7
 se 831NN ~~4/10~~ y = 16 75 n' 1.1' 33
 y = 10 75 n' 1 se 33 (7, 17, 19, 21)
 (5, 15) se 831NN

question 16

(a)

$$X_1: \bar{X}_1 = \frac{5+15+15+25}{4} = 15$$

$$S^2 = \frac{(5-15)^2 + (15-15)^2 + (15-15)^2 + (25-15)^2}{4} = 50$$

$$S^2(X_1) = 50$$

$$X_2: \bar{X}_2 = \frac{10+30+50+30}{4} = 30$$

$$S^2 = \frac{(10-30)^2 + (30-30)^2 + (50-30)^2 + (30-30)^2}{4} = 200$$

$$S^2(X_2) = 200$$

$$X_3: \bar{X}_3 = \frac{5+15+15+5}{4} = 10$$

$$S^2 = \frac{(5-10)^2 + (15-10)^2 + (15-10)^2 + (5-10)^2}{4} = 25$$

$$S^2(X_3) = 25$$

(b)

$$\text{Total Variance} = 50 + 200 + 25 = 275$$

$$\boxed{\text{Total} = 275}$$

(c) $PC_1 (0.6, 0.8, 0)$

driver	$0.6X_1 + 0.8X_2 + 0X_3$	Projection
1	$0.6(5) + 0.8(10)$	11
2	$0.6(15) + 0.8(30)$	33
3	$0.6(15) + 0.8(50)$	49
4	$0.6(25) + 0.8(30)$	39

$$\bar{X} = \frac{11 + 33 + 49 + 39}{4} = 33$$

$$S^2_{PC_1} = \frac{(11-33)^2 + (33-33)^2 + (49-33)^2 + (39-33)^2}{4} = \boxed{194}$$

$$\boxed{\text{ratio} = \frac{194}{275} = 0.705}$$